

Member ID: _____

Time: _____

Rank: _____



MANAGERIAL ACCOUNTING (135)

NATIONAL 2022

CONCEPT KNOWLEDGE:

Multiple Choice (15 @ 2 points each) _____ (30 points)

APPLICATION KNOWLEDGE:

Short Answer _____ (51 points)

Problem 1: Incremental Analysis _____ (25 points)

Problem 2: Capital Investment Analysis _____ (35 points)

TOTAL POINTS _____ ***(141 points)***

Test Time: 90 minutes

GENERAL GUIDELINES:

Failure to adhere to any of the following rules will result in disqualification:

1. Member must hand in this test booklet and all printouts if any. Failure to do so will result in disqualification.
2. No equipment, supplies, or materials other than those specified for this event are allowed in the testing area. No previous BPA tests and/or sample tests (handwritten, photocopied, or keyed) are allowed in the testing area.
3. Electronic devices will be monitored according to ACT standards.

EXAM GUIDELINES:

You have been hired as a Financial Assistant and will be keeping the accounting records for Digital Solutions, located at 700 Morse Road, Suite 201, Columbus, Ohio 43214. Digital Solutions provides accounting and other financial services for clients. You will complete jobs for Digital Solutions' own accounting records, as well as for clients.

You will have 90 minutes to complete your work. The test is divided into two parts: concept knowledge and application knowledge.

Your name and/or school name should *not* appear on any work you submit for grading. Write your Member ID in the provided space. Staple all pages in order before you turn in your test.

Assumptions to make when taking this assessment:

- Round all calculations to two decimal places at the final step.
- Round all percentages to one decimal place.
- Use 360 days for interest calculations.

Multiple Choice: Select the letter of the response that *best* answers the question.

1. Which of the following best describes a standard?
 - a. Budgeted amount for total product cost
 - b. A sales price, cost, or quantity that is expected under normal conditions
 - c. Costs incurred to produce a product
 - d. Actual sales price, cost, or quantity
2. The purchasing manager for Dunder Mifflin was able to bring down the cost of direct materials by purchasing direct materials of a slightly lower grade quality than the company had used previously. The lower grade of direct materials, however, meant a higher defect rate on the assembly line and higher wastage of direct materials during production, which in turn lowered operating income. This would have led to a(n) _____.
 - a. favorable direct labor cost variance
 - b. unfavorable direct materials cost variance
 - c. unfavorable direct materials efficiency variance
 - d. favorable direct labor efficiency variance
3. Assuming actual sales volume is 11,000 units and planned volume is 10,000 units, the sales activity variance _____.
 - a. is 1,000 units favorable.
 - b. is 1,000 units unfavorable
 - c. cannot be determined without additional information
 - d. activity variances are neither favorable nor unfavorable
4. The balanced scorecard _____.
 - a. incorporates financial and non-financial measures in an integrated system
 - b. is based on financial measures
 - c. is based on non-financial measures
 - d. does not use financial or non-financial measures
5. Companies in which owners or top executives make all of the planning, directing and controlling decisions are _____ companies.
 - a. centralized
 - b. formalized
 - c. segmented
 - d. decentralized
6. Michael Scott, one of the managers of a multi-national company, is responsible for generating revenues and controlling costs in order to increase the operating income of his division. However, he is not concerned about investment-related decisions. Michael is most likely to be the manager of a(n) _____.
 - a. profit center
 - b. revenue center
 - c. cost center
 - d. investment center

7. Rose Apothecary has several divisions that are investment centers. Data for the Skincare Division and the Apparel Division are shown here:

	Skincare Division	Apparel Division
Operating income	\$70,000	\$40,000
Total assets at Jan. 1	\$680,000	\$215,000
Total assets at Dec .31	\$780,000	\$210,000

Which of the following statements would be the most meaningful interpretation of this data?

- The Skincare Division shows a more efficient use of assets than the Apparel Division because it has a higher operating income.
 - The Skincare Division uses its assets less efficiently than the Apparel Division does because it has a lower return on investment.
 - The Skincare Division is financially more successful than the Apparel Division because it shows an increase in assets.
 - The Skincare Division's performance is better than that of the Apparel Division because it has higher assets.
8. Costs that do not differ between alternatives are _____.
- considered irrelevant to the decision
 - considered opportunity costs
 - relevant to the decision
 - important only if they represent a material dollar amount
9. _____ is (are) expected future data that differ among alternatives.
- Irrelevant costs
 - Relevant costs
 - Relevant information
 - Sunk costs
10. Which of the following is always irrelevant to short-term operating decisions?
- Sunk cost
 - Differential cost
 - Opportunity cost
 - Relevant cost
11. An opportunity cost is _____.
- the cost incurred to gain the opportunity to make a sale
 - the benefit gained by choosing a certain course of action
 - costs that have been incurred in the past
 - the benefit given up by choosing an alternative course of action

12. _____ is the amount by which actual sales of a company exceeds its break-even sales level
- Contribution margin
 - Margin of safety
 - Operating leverage
 - Relevant range
13. Which of the following best describes the term "capital rationing"?
- A process of controlling operating costs when adequate funds are not available.
 - A method that shows the effect of an investment on a company's accrual-based income.
 - A process of ranking and choosing among alternative capital investments based on the availability of funds.
 - A method of determining the period within which the cash invested is recouped.
14. Which of the following best describes the internal rate of return?
- The ratio of average annual income to average amount invested.
 - Interest rate that makes the net present value of the investment equal to zero.
 - The rate at which the profitability of an investment increases.
 - Discount rate that is used to evaluate funds borrowed from a lender for profitability.
15. All other things being equal, the _____ the expected rate of return, the more attractive the investment opportunity
- lower
 - higher
 - expected rate of return has no effect on the investment opportunity
 - impossible to determine based on the information given

Short Answer (51 Points)

1. Stark Industries has expected sales ranging from \$170,000 to \$200,000. Variable costs and their percentage relationships are: Sales commissions 5%, Advertising 4%, Traveling 3%, and Hair Supplies 8%. Fixed expenses are Salaries \$34,000, Depreciation \$7,000, and Insurance \$1,000.

If sales are \$180,000, what are total costs?	
---	--

Actual expenses for Stark Industries are as follows:

Variable Expense		Fixed Expenses	
Sales Commissions	\$9,200	Salaries	\$34,000
Advertising	7,000	Depreciation	7,000
Travel	5,100	Insurance	1,000
Hair Supplies	16,800		

If sales are \$180,000 what is the budget variance for total expenses?	
Is it Favorable or Unfavorable?	

2. Acme Corp. has budgeted three hours of direct labor per anvil, at a standard cost of \$17 per hour. During August, laborers actually worked 189 hours completing 84 anvils. All 84 anvils actually produced were sold. Acme paid the laborers \$18 per hour.

Calculate the following variances and identify if each was favorable (F) or unfavorable (U).

	Variance	Favorable or Unfavorable
Labor Rate Variance		
Labor Efficiency Variance		
Total Labor Spending Variance		

3. Bluth Industries manufactures kiosks for shopping malls. During the second quarter of the year, the company manufactured 3,500 kiosks, using 22,500 pounds of plastic. The plastic cost the company \$117,100. According to the standard cost, each kiosk should require 6 pounds of plastic at a cost of \$8 per pound.

Calculate the following variances and identify if each was favorable (F) or unfavorable (U).

	Variance	Favorable or Unfavorable
Materials Price Variance		
Materials Quantity Variance		
Total Materials Spending Variance		

4. Central Perk provides the following financial information:

Income from operations	\$200,000
Interest expense	50,000
Gains/(losses) on sale of equipment	3,000
Net income	147,000
Total assets at Jan. 1	2,900,000
Total assets at Dec .31	3,300,000
Target Rate of Return	8%

Calculate return on investment	
Calculate the Residual Income	

Problem 1 (25 Points)

- A. It costs Wonka Inc. \$45 per unit (\$27 variable and \$18 fixed) to make a product that normally sells for \$80. A wholesaler is offering to buy 5,000 units at \$30 each. Wonka will incur special shipping costs of \$1 per unit.

Assuming that Wonka has excess capacity how much income (loss) would Wonka realize by accepting this special order?	
---	--

- B. Duff's Corp. incurs unit costs of \$13 (\$8 variable & \$5 fixed) in making a part for its product. A supplier has offered to make 10,000 of the parts at \$7.30 per unit. If the offer is accepted, Duff's will save all variable costs but no fixed costs.

How much money will Duff's save (lose) by buying the part?	
--	--

- C. Wayne Industries makes an unassembled product that it currently sells for \$40. Production costs are \$28. Wayne is considering assembling the product and selling it for \$68. The cost to assemble the product is \$12.

Should Wayne sell <i>before assembly or process further</i> ?	
If Wayne decides to process further, net income will increase (decrease) by how much per unit?	

- D. Spacely Space Sprockets manufactures three types of sprockets – the Cosmo, the Astro, and the Rosie. The company is deciding whether to drop the Rosie product line because it has an operating loss.

	Cosmo	Astro	Rosie
Sales revenue	\$80,000	\$40,000	\$30,000
Variable costs	(40,000)	(15,000)	(12,000)
Contribution margin	\$40,000	\$25,000	\$18,000
Fixed costs	(15,000)	(5,000)	(24,000)
Operating income (loss)	\$25,000	\$20,000	\$(6,000)

Assuming fixed costs are unavoidable, if Spacely drops the Rosie product line and does not replace it, what effect will this have on operating income?	
--	--

Problem 2 (35 Points)

Grey-Sloan Memorial is evaluating two possible investments in depreciable plant assets. The company uses the straight-line method of depreciation. The following information is available:

	Investment A	Investment B
Initial capital investment	\$108,000	\$151,000
Estimated useful life	10 years	10 years
Estimated residual value	0	\$23,000
Estimated annual cash inflow for 10 years	\$30,000	\$45,000
Estimated annual cash outflow for 10 years	\$5,000	\$7,000
Required overhaul	\$20,000 at the end of 5 years	\$15,000 every 4 years
Required rate of return	10%	14%

For each investment, calculate the following:

	Investment A	Investment B
Payback Period		
Accounting Rate of Return/Simple Rate of Return		
Net Present Value		

Which investment is preferable? _____